7 Reference

FULL

Jun's Garden: the handover, in one page

The details of every change are in `CHANGES.md` (what, why, and the effect on the program), the story in plain words is in `JOURNAL.md`, and every step is in the git history. This page is the short version.

The public code: https://github.com/netnovaops-training/garden-for-all. Anyone can read and check it there.

The scope

The goal: hand the garden website junsgarden.zapto.org to its owner, Jun, so he can run it on his own computer, by himself: fully transparent, free, public, and with no way in for anyone but him.

In scope:

- The live website (Rust), the garden tools (`/beds`), the optional wiki, and the Android app's source.
- An installer that sets everything up and explains every step: `garden.sh` on Ubuntu (14 resumable steps, EASY or CUSTOM, every option in a menu), and the Garden Bridge on Windows (a guided installer with Next and Back).
- Guides in English and Korean, each with a quick version, as PDFs and as a web handbook that opens from the disk, and a welcome page with every guide.
- Backups, going online with his own web name, a sign-in page, remote access, and the tests that prove each part works.

Out of scope, on purpose:

- The old PHP site, the old household server and its services, and every login, key and account of the author.

- The garden's own data (plants database, photos, wiki pages): a separate, private download for Jun, not in this repository.
- Ideas for a later version (not in this one): a starter garden for new people (free-to-use plants for their area, their own photos, and credit to the original Jun's Garden code), and custom addresses and ports for the front door.

Enhancements

- **The install does the work.** Every program is installed by the install itself, with each step explained (what, why, the official page) and its commands shown before they run, a timer on each step, and a full log. It can stop and carry on where it stopped.
- **Three levels to choose from**, so Jun can plan his time: Easy (beer, about 1-2 hours), Recommended (wine, about 2-4 hours over several days) and Full (champagne, about 3-5 hours), each with its chance of a frustrating moment (3, 5 and 8 out of 10). Guide 23 is the full setup, part by part.
- **Windows:** the Garden Bridge installer; it offers to move the garden folder out of OneDrive; a welcome page opens when the install ends.
- **Ownership:** no built-in account; Jun's own keys, his own No-IP name, his own Google client ID; the Website name swap changes the web name in every file at once.
- **Email for "forgot password":** through Gmail with an app password, with the two Google pages shown where it asks, and a second "are you sure?" before skipping.
- **Backups, 3-2-1:** every night, to up to three network places (NAS, a second NAS, a PC's shared folder), Google Drive, and a Windows folder Jun picks (a USB drive, Google Drive for desktop, a NAS).
- **Remote access:** SSH with keys, WinSCP for files between Linux and Windows, Termius for a terminal on the phone, and Tailscale (menu M, 17) to reach the computer from anywhere with no port opened.
- **Guides:** 30 in English and 28 in Korean, plus 12 project and automation guides; a beginner's bash walkthrough; 43 pictures taken from the real program (and 4 diagrams),

with sample names and rings on what to click; PDFs with a colour per step; a web handbook with a step > guide > topic tree for when Jun is stuck.

- **Korean** on screen too: the menu speaks English, Korean or both (every one of its sentences is translated).
- **Rename and move, safely** (menu M, 18; guide 26): a file or folder of the website gets Jun's own name, and every place that uses it follows (code, pages, photo lists, database), with a preview, a backup and a check. Big folders move to another disk; the whole garden folder moves too. The guides keep the first names, and `logs/renames.tsv` lists what is called what now.
- **An on / off switch** (menu M, 14): everything off without deleting anything, on again even months later; a check-up that restarts a stuck program (also every 6 hours if wanted); a clean-up that removes only Docker's leftovers older than 7 days, after asking.
- **Windows: two ways to run Docker:** Docker Desktop (the easy way, with buttons) or Docker inside Ubuntu (the hands-on way), chosen in the installer. The installer can find the garden computer at home for you ("Find it for me"), and the menu finds it again by its network card if its address changes.
- **No AI inside:** the menus, the installer and the app are plain scripts and programs, each doing one job, so every step can be read.

Fixes

Testing every option on Ubuntu, on Windows (the Garden Bridge, downloading from Google Drive as Jun will) and on a Raspberry Pi 5 found **52 problems**, all fixed and each tested again (`CHANGES.md`, Parts 2, 5 and 6). A few examples:

- The firewall step could have locked out SSH on a computer whose SSH uses another port; it now keeps the real port open, from home only.
- Once Docker was installed, "home Wi-Fi only" let only Docker's own network in.
- "Remove everything" left the programs behind; it now removes all it lists.
- Yes/no checks could answer a wrong "no" when a program's output was long.

- A Korean Ubuntu's folders (다운로드, 바탕화면) were not searched for the downloads.
- Moving the garden folder out of OneDrive would have broken the scheduled backup copy.
- A clean-up command in the guides could have deleted other Docker projects' stopped programs on a shared Docker; the new clean-up touches only the garden's leftovers.

Security

A separate review read every file and the whole history before publishing. The key answer: **someone who reads this code, even knowing Jun's name and email, cannot sign in as Jun or anyone else.** No passwords, keys or accounts are in the code; every install makes its own secrets; only the front door (ports 80 and 443) faces the internet. Ten improvements came from the review (`CHANGES.md`, Part 4).

Checked before release

- Website: 64 of 64 tests pass. Shell scripts: shellcheck clean (18 files).
- The menu's Korean: every sentence translated. Korean guides: 0 differences from the English in links, commands and pictures.
- 219 outside links checked, 0 broken. Secret scan (gitleaks, semgrep): clean.
- Install tested: Ubuntu (Windows' own Ubuntu), the Windows path from Google Drive, and a Raspberry Pi 5 (arm64; the build-and-start part, 8 minutes, 0 fails).

한국어 요약

준의 정원 웹사이트를 주인인 준에게 넘겨서, 준이 자기 컴퓨터에서 혼자 운영할 수 있게 한 프로젝트예요. 투명하고, 무료이고, 공개되어 있고, 준 말고는 아무도 들어올 수 없어요.

- **범위:** 웹사이트(Rust), 정원 도구, 선택으로 위키, Android 앱 소스, 설치 프로그램 (Ubuntu의 `garden.sh`, Windows의 Garden Bridge), 영어·한국어 안내서(PDF와 웹 안내서), 백업,

인터넷에 올리기, 원격 접속, 그리고 모든 테스트. 예전 서버와 그 계정·키, 그리고 정원의 데이터(따로 내려받는 파일)는 포함하지 않아요.

- **좋아진 것:** 설치가 모든 일을 하고 단계마다 설명해요. 세 단계(맥주·와인·샴페인)로 시간을 계획할 수 있어요. 이메일 비밀번호 찾기, 3-2-1 백업, Tailscale로 어디서든 나만 접속. 파일과 폴더에 내 이름 붙이기(쓰는 곳이 모두 따라와요), 켜기 / 끄기 스위치와 점검, Windows에서는 Docker Desktop(쉬운 방법)이나 Ubuntu 안의 Docker 중에서 고르기(안내서 26). 메뉴와 앱에는 AI가 없어요: 하나에 한 가지 일만 하는 평범한 스크립트예요.
- **고친 것:** 테스트에서 찾은 문제 52개를 모두 고치고 다시 확인했어요(`CHANGES.md`).
- **나중 버전을 위한 생각:** 새로 시작하는 사람을 위한 시작 정원(자유롭게 쓸 수 있는 식물, 자기 사진, 원래 Jun's Garden 코드에 대한 감사 표시), 현관문 주소와 포트 바꾸기.
- **보안:** 이 코드를 읽어도, 준의 이름과 이메일을 알아도, 준으로 로그인할 수 없어요.